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An image quality-aware approach with adaptive scattering coefficients for single image dehazing

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Abstract

Most conventional dehazing methods obtain quality results by solving atmospheric scattering model (ASM) using acquired variables (i.e., global atmospheric light and transmission map). Prior-based strategies have made significant achievements in this task. Nonetheless, they usually obtain unrealistic dehazed images since strong assumptions can barely suit all circumstances. In this paper, we propose a novel image dehazing method with adaptive scattering coefficients to realize visual-friendly and quality-orientated restoration. Specifically, a regional rank-based technique is applied to find the most likely atmospheric light candidate. And then, different from previous image dehazing methods that rely on haze-relevant priors to estimate a transmission map, we develop an image quality-aware approach, together with a dynamic scattering coefficient. In this phase, an optimization function constrained by the image quality-aware indicators is designed to compute the scattering coefficient or transmission. The Fibonacci algorithm is further employed to solve this optimization problem. The proposed method produces high-quality results and exhibits favorable quantitative and qualitative performance compared to related methods.

Keywords Image dehazing · Atmospheric scattering model · Adaptive scattering coefficient

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1 Introduction

Images taken under hazy conditions are commonly suffered from low contrast and dim color, owing to the interference from floating particles in the atmosphere. This degradation can impair the performance of computer vision applications that rely on visual features, including object classification [20] and image detection [8, 37]. Thus, an effective image dehazing solution is desirable for eliminating unfavorable effects and reconstructing reduced visual information.

Studies show that the most intuitive strategy is to globally or locally improve the contrast of an image by traditional enhancement techniques (e.g., Histogram Equalization (HE) [26], Retinex [12], and image fusion-based methods [38]). These methods can enhance image contrast to a certain extent, but the visual quality of the final enhanced images is quite limited due to ignoring the precise properties of image haze theory. Recently, atmospheric scattering model (ASM) based restoration techniques have received a great deal of attention. Here, we provide a brief overview of the atmospheric scattering model, illustrated in Fig. 1, which describes the interaction of environmental illumination with haze. As we can see, atmospheric particles scatter photons away from their original trajectory, leading to a loss of energy in the incident light. Only a fraction of photons from the original scene can reach the optical imaging sensor. The ASM quantifies the radiance contributed by scene and illumination source, which can be expressed as:

$$I^c(x) = A^c \rho^c(x) t(x) + A^c(1 - t(x)) \tag{1}$$

where $c \in \{R, G, B\}$ denotes the color channel index, x denotes the position of a pixel, $I^c(x)$ refers to the observed hazy image, $\rho^c(x)$ is scene albedo (i.e., clear scene), A^c is the atmospheric light, and $t(x)$ is the medium transmission that can be further expressed as:

$$t(x) = e^{-\beta d(x)} \tag{2}$$

where β is the atmospheric scattering coefficient, and $d(x)$ is the scene depth. The core idea of image dehazing is to estimate a clear image $\rho(x)$ from the taken target input $I(x)$. Because there are more than two derived parameters (i.e., A^c, β, and $d(x)$) in the ASM, we cannot learn them without extra information [39]. Therefore, some researchers have introduced numerous priors as additional information to pinpoint a reasonable solution for $\rho(x)$. These prior-based

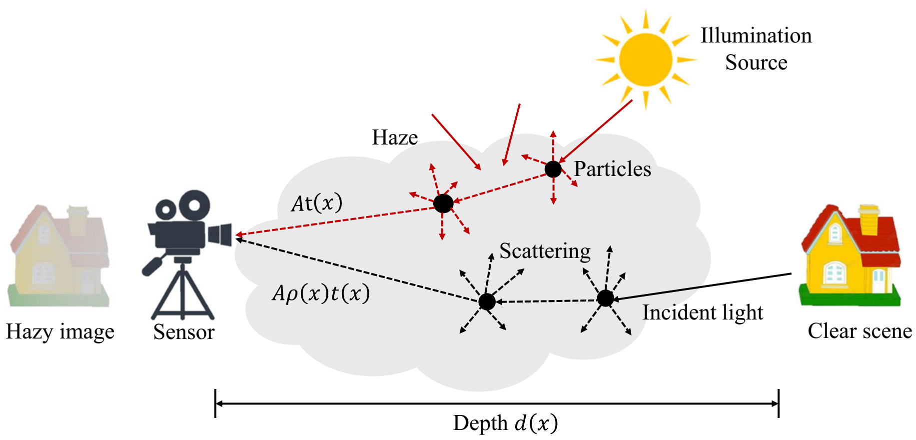

Fig. 1 Diagram of light scattering caused by haze

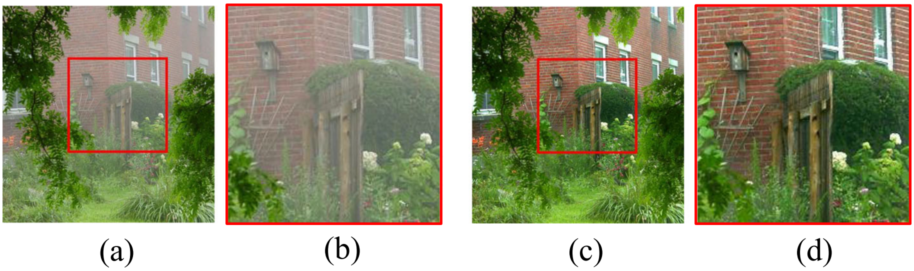

Fig. 2 Example of image dehazing produced by the proposed method. (a) Original image. (b) The local region in the red box of (a). (c) Dehazed image. (d) The local region in the red box of (c)

methods may be effective in certain cases, but they are not universally applicable because constructing robust statistical priors is challenging and frequently does not offer satisfactory generalization capabilities.

In this paper, we propose an image quality-aware approach with adaptive scattering coefficients to improve the dehazing performance. Concretely, we first present a regional rank-based technique to calculate the atmospheric light. Interestingly, we hope that the dehazed results can possess desired properties of image quality, such as high image contrast and rich texture information. To this end, image quality-aware feature indicators are incorporated into the scattering or transmission estimation step. At last, the restored image is obtained by inversion the hazy imaging model. The proposed method can effectively removes haze and improves the visual quality of images, as shown in Fig. 2. In summary, the specific contributions of this paper are highlighted as follows.

- We propose an image quality-aware dehazing method aimed at achieving effective haze removal while reducing halo artifacts in large smooth regions and improving color fidelity. Extensive experiments show that the proposed method introduces excellent robustness and flexibility, when compared with the state-of-art methods.
- A regional rank-based solution is adopted, which employs the main idea of seeking an ideal region with the lowest score and using it to estimate the atmospheric light. This strategy effectively suppresses the interference of white objects on atmospheric light estimation and avoids the problem of dark dehazed results caused by overestimation of atmospheric light.
- The proposed method designs an optimization function by fully leveraging the latent properties of image. Then, the scattering coefficient or transmission is obtained by solving the optimization function using the Fibonacci algorithm. Thanks to this solution help us to eliminate statistical priors on the ASM for transmission estimation, our dehazing method can achieve improvements in visual image quality.

2 Related work

Significant progress has been achieved in the field of image dehazing in the past few decades. Current methods can generally be categorized into the following groups: fusion-based method, learning-based method, and model-based method.

2.1 Fusion-based method

The fusion-based image dehazing methods combine multiple images according to different weights to dehaze the image. For example, Ancuti et al. [3] fused two images derived through white balance and contrast enhancement in a multiscale manner to achieve the goal of dehazing. Li et al. [32] derived the features of dark channel, sharpness, and saliency from two enhanced images as fusion weight maps. And they used a multiscale pyramid fusion strategy to restore the hazy images. However, the dehazed results obtained using these methods cannot contain the most details in the original image scene. Galdran et al. [17] first obtain articially underexposed images through gamma-correction operations. They then combined a set of multiple exposure images into an output using a multiscale Laplacian blending scheme to enhance the global exposure quality of the image. However, local feature enhancement remains uncertain. Zhu et al. [45] also used gamma correction to obtain four underexposed images from the blurry images. They then analyzed global and local exposures to obtain a pixel-based exposure weight map for image fusion. Similarly, [28]-[43] achieve image restoration by optimizing local and global exposure quality as well.

2.2 Learning-based method

Learning-based methods have recently witnessed significant progress in image dehazing. Chen et al. [41] designed PMS-Net, a network that determines patch sizes suitable for each pixel to generate a patch map, allowing accurate estimation of atmospheric light and transmission. Cai et al. [6] proposed DehazeNet, an end-to-end learning method that estimates the medium transmission of a single hazy image based on established image priors. However, it lost effectiveness in dense scenes. Huang et al. [21] proposed an end-to-end network that uses transition convolutional layers to enhance structural information for haze removal. Wu et al. [5] presented a model for image dehazing using an autoencoder-like dehazing network and contrastive regularization (AECR-Net). Li et al. [25] proposed a fully-functional image restoration network, named AirNet, which can recover images from multiple damages.

Some methods tend to incorporate traditional prior knowledge to construct networks that achieve good dehazing effects in real-world scenarios. For instance, Shao et al. [15] proposed a domain-adaptive paradigm for image dehazing by embedding the dark channel prior (DCP) into their learning-based approach and constructing a bidirectional translation network. Dong et al. [13] utilized high-frequency and low-frequency information as additional priors for generating dehazed images by developing a novel fusion discriminator. Chen et al. [2] proposed a new paradigm of multi-prior guided dehazing network that considered the complementarity of multiple priors, including the dark channel loss, the bright channel loss, and the contrast-limited adaptive histogram equalization loss, to guide unsupervised training for obtaining high-contrast images.

2.3 Vision transformers

Recently, vision transformers are getting increasing attention for vision tasks, and their excellent performance has been demonstrated. Their outstanding performance has shown in several vision tasks: image recognition [7, 29], image segmentation [24] and object detection [11]. For example, Chen et al. [7] proposed a novel module, named the spatial and semantic transformers (SST). The module comprised two independent transformers, to simultaneously capture these two aspect correlations, respectively. Zhao et al. [29] proposed a dual relation learning framework based on the transformer, for multi-label recognition tasks. Karimi et al. [24] proposed a transformer network architecture without any convolution operations. This deep neural network architecture is capable of more accurate segmentation than fully convolutional networks (FCNs). Chen et al. [11] proposed the adaptive image transformer

(AIT) module, which can well predict the class similarity between the target image and the query image block. The result showed that the effectiveness of the one-shot object detection (OSD) task has been enhanced.

In addition to the applications mentioned above, new versions of vision transformers have been developed and used in image enhancement [42]. For example, Zamir et al. proposed an efficient Transformer model that captures long-range interactions between pixels through critical designs in building blocks, and is also applicable to large-scale images. Peng et al. [14] first introduced the converter model into the underwater image enhancement (UIE) task and proposed a U-shaped transformer network. Zhao et al. [36] introduced a new transformer block to reduce the deficiencies of CNNs-based dehazing networks. The new transformer block incorporated within the dehazing networks, dubbed the HyLoG-ViT, can capture local and global dependencies.

2.4 Model-based method

Hazy image restoration methods calculate atmospheric parameters (atmospheric light and transmission) based on the optical model. To solve the ill-posed problem, prior knowledge has been proposed. For example, He et al. [18] came up with an image haze removal method of a dark channel prior (DCP) through the analysis of a large number of outdoor haze-free images. Currently, the DCP is still widely used which can be expressed as:

$$I^{\text{dark}}(x) = \min_{y \in \Omega(x)} \left(\min_{c \in \{r,g,b\}} I^c(y) \right) \tag{3}$$

where $\Omega(x)$ is a local patch centered at x, $c \in \{r, g, b\}$ is the color channel index. The top 0.1% brightest pixels in $I^{\text{dark}}(x)$ were chosen and these pixels are mapped to the input image. Among these pixels, the pixel with the highest intensity was calculated as the atmospheric light. It can be described as:

$$A^c = I^c \left(\arg\min_{x \in p_{0.1\%}} \sum_{c \in \{R,G,B\}} I^c(x) \right) \tag{4}$$

where $p_{0.1\%}$ is the set of positions of top 0.1% brightest pixels in $I^{\text{dark}}(x)$.

The DCP assumes that the transmission is constant in the local patch. Then, the minimum operator is applied on (1), that is:

$$\min_{y \in \Omega(x)} \left\{ \min_{c \in \{r,g,b\}} I^c(y) \right\} = A^c(1 - t(x)) + \min_{y \in \Omega(x)} \left\{ \min_{c \in \{r,g,b\}} J^c(y) \right\} t(x) \tag{5}$$

For a haze-free outdoor image, $J^{\text{dark}} \to 0$, that is:

$$J^{\text{dark}}(x) = \min_{y \in \Omega(x)} \left(\min_{c \in \{r,g,b\}} J^c(y) \right) = 0 \tag{6}$$

Since $J^{\text{dark}} \to 0$ and $A^c > 0$, according to (3), the transmission can be expressed as:

$$t(x) = 1 - \min_{y \in \Omega(x)} \left\{ \min_{c \in \{r,g,b\}} \frac{I^c(y)}{A^c} \right\} \tag{7}$$

The restored image can be expressed by putting $I^c(x)$, $A^c(x)$ and $t(x)$ into (1). The haze-free image can be written as:

$$J^c(x) = \frac{I^c - A^c}{\max(t(x), t_0)} + A^c \tag{8}$$

where $t_0 = 0.1$.

The DCP method assumes a locally constant transmission in a rectangular patch of a fixed size, which can result in halo artifacts near depth discontinuities and lead to suboptimal restoration outcomes. To refine the transmission, the soft matting (SM) [27] and guided filtering (GF) [19] are used. It is worth noting that the SM can eliminate the halo artifacts very well, but its time complexity is greatly increased. The time complexity of the GF is small, but its recovered image still has a certain degree of haze in the edge mutation region. Zhang et al. [9] proposed the maximum reflectance prior that utilizes the high intensity of each color channel in certain pixels of a haze-free image captured during the daytime, and estimates ambient illumination based on this prior. Fattal et al. [16] developed a local formation model for image dehazing, which uses color lines to estimate the transmission. Zhu et al. [44] developed a linear model for depth of field using the color attenuation prior (CAP) and obtained the model parameters through supervised learning. The method solves for the transmission using the scattering coefficient and scene depth. To reduce the number of parameters, a constant value of 1 is typically used for the scattering coefficient in CAP. Berman et al. [4] proposed a color-based nonlocal-wise method for estimating transmission and atmospheric light, which is effective but suffers from oversaturation. [22] proposed a dehazing method with gamma-corrected prior (IDGCP). They first used gamma-corrected prior to obtain homogeneous virtual transformations of hazy images. Then, they calculated scene depth ratios and scene albedos with the help of the atmospheric scattering model. To avoid the effect of recovered images being dim, Ju et al. [23] incorporated a light absorption coefficient into the optical model, which obtained a new dehazing model (EASM) to avoid the effect of recovered images being dim. Based on EASM and the grayscale world assumption, they proposed an image dehazing method called IDE.

To overcome the flaws in transmission map estimation that result in unnatural output, we propose a novel image dehazing method. Further details will be provided in Sect. 3.

3 Proposed method

In this section, we introduce a new image dehazing method, in which atmospheric light and transmission estimation are provided. First, we employ a regional rank-based method to estimate the atmospheric light. Then, the scattering or transmission is estimated by solving an optimization function, which is constrained by the image quality-aware feature indicator. The specific steps of this method are as follows.

3.1 Atmospheric light estimation

As discussed in Sect. 2.4, the DCP method applied a global solution to estimate the atmospheric light A^c. When facing hazy images with large bright objects, DCP will not work well (see Fig. 4). To ameliorate this issue, we present a novel strategy to estimate the atmospheric light.

Regions that involve bright objects typically introduce a lower mean pixel value in the dark channel compared to regions controlled by genuine atmospheric light. Therefore, it is necessary to eliminate the adverse effects of white objects on the dark channel. Interestingly, we note that the color tone of a hazy image has been changed by a depth-dependent additive factor, compared with the real scene. Objects in comparison to regions of atmospheric light often show more variations within R-G-B component of the target input and embody more textural information. To this end, our goal is to seek an ideal region that is most haze-opaque and free of objects. Following this intuition, we use two features f_1 and f_2 derived from the hazy image I to estimate A^c. For clarity, the entire process of atmospheric light value estimation for the proposed method is outlined in algorithm 1.

Algorithm 1 Atmospheric Light value A^c Estimation Algorithm

Input: original image I, segmentation block numbers $N = 500$
Do the job:
1. Compute the pixel-based dark map(C^d) and gradient map (C^g) from the original image I
2. Apply the SLIC algorithm to segment the original image I into N patches and obtain the Mask
3. Apply the Mask to the pixel-based dark map(C^d) and gradient map (C^g) into N patches
4. For each patch on pixel-based dark map(C^d), do
   Compute pixel-based dark map(f_1) using (9) and (10)
   End for
5. For each patch on the gradient map (C^g), do
   Compute the mean gradient map(f_2) using (11) and (12)
   End for
6. Normalize f_1 and f_2, then add them together to obtain the fused decision map F
7. Obtain the index of the atmospheric light value region by $k = \arg\min_i (F_i)$
8. Compute the atmospheric light value A^c from the block at index k using (13)

Output: Atmospheric light value A^c

Taken I as an input, it is first divided into non-overlapping N image patches (Mask) by using SLIC (Simple Linear Iterative Clustering) [1]. The Mask of the input image is shown in Fig. 3(b). Then, pixel-based dark channel map (C^d) and gradient map (C^g) are obtained. Their mathematical formulas will be described in detail in (10) and (12). Next, we naturally use the Mask on C^d and C^g (see Figs. 3(c) and 3(d)) to guarantee the segmented results are consistent, respectively. The first feature is designed to make use of the dark channel, denoted as the revised pixel-based dark (RPD) map (see Fig. 3(e)). RPD map (f_1) of the i^{th} block is

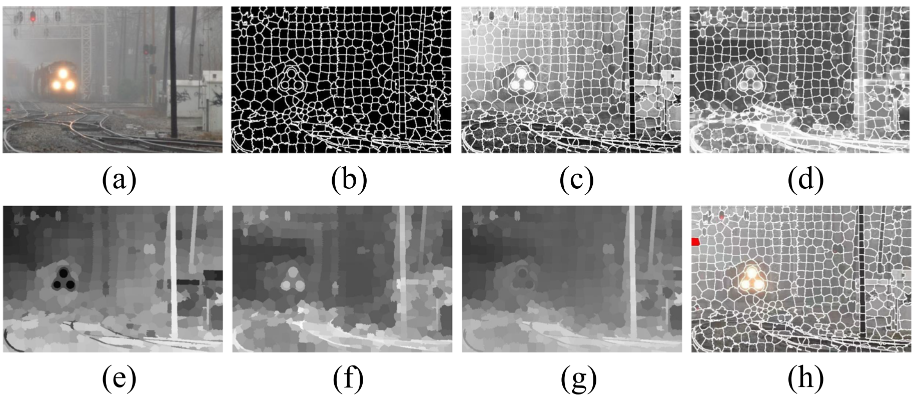

Fig. 3 Example of each step of atmospheric light estimation. (a) The input image. (b) The Mask of the input image. (c) (d) Segmented maps obtained by applying the Mask to the pixel-based dark channel map and gradient map, respectively. Atmospheric light estimation features (normalized) using (e) the feature map f_1 and (f) the feature map f_2. (g) The fusion decision map F. (h) The position of atmospheric light is the red region

defined as:

$$f_{1_i} = 1 - \mu_i^d \in [0, 1] \tag{9}$$

where

$$\mu_i^d = \frac{1}{|\Omega_i|} \sum_{x \in \Omega_i} \left(C^d (x) \right) \tag{10}$$

where $i = (1, 2, \ldots, N)$, $C^d (x) = \min_{c \in \{R,G,B\}} I^c(x)$ is the pixel-based dark map, μ_i is the mean of the i^{th} region of the C^d, and $|\Omega_i|$ represents the number of pixels in Ω_i.

The second feature is used to reveal image edges, denoted as mean gradient (MG) map (see Fig. 3(f)). MG map (f_2) of the i^{th} block is defined as:

$$f_{2_i} = \frac{1}{|\Omega_i|} \sum_{x \in \Omega_i} \left(C_i^g (x) \right) \in [0, 1] \tag{11}$$

where

$$C_i^g (x) = \frac{1}{n} \sum_{j=1}^{n} \left| I_g (x) - G^{r_j, r_j} (x) \right| \tag{12}$$

where $C_i^g (x)$ is the i^{th} block of the gradient map. I_g is the grayscale version of I^c. G^{r_j,r_j} is the input image filtered by a $r_j \times r_j$ spatial Gaussian filter with variance r_j, $r_j = 2^j n + 1$ and n is set to 3.

Two normalized maps f_1 and f_2 are added together to obtain $F_i = f_{1_i} + f_{2_i}$, denoted as the fused decision (FD) map (see Fig. 3(g)). Lower F_i values indicate a higher probability of atmospheric light. Thus, we select $k = \arg\min_i (F_i)$ as the candidate region. For example, in Fig. 3(h), the red region is selected. Finally, A^c can be defined as:

$$A^c = \frac{1}{|\Omega_k|} \sum_{x \in \Omega_k} I^c(x) \tag{13}$$

Fig. 4 illustrates the comparison results and atmospheric light positions obtained by the DCP method and the proposed method. The expected atmospheric light regions estimated by the DCP method are located in the white car and train light (the red regions in Figs. 4(a) and 4(c)). The results of DCP show relatively low brightness due to the overestimation of the atmospheric light. To address this issue, we introduce a novel strategy for estimating the atmospheric light. As shown in Figs. 4(d) and 4(h), the proposed method achieves a more natural dehazing effect and significantly enhances image contrast. To further demonstrate the effectiveness of the proposed method, we annotate the fog aware density evaluator (FADE) [10] and hue restoration ability metric (HRAM) [30] in Fig. 4 to evaluate the dehazing and color restoration abilities of the resulting images. By comparing the metrics, it can be observed that the proposed method outperforms the DCP method, highlighting the competitiveness of our approach.

3.2 Transmission map estimation

Haze can cause colors to fade and distort in the image. Due to the scattering effect, the colors of different objects can be mixed together, making it difficult to distinguish them. By estimating the scene albedo(i.e., haze-free image), the true color information of objects in the image can be restored while removing the haze, improving the visual quality of the image.

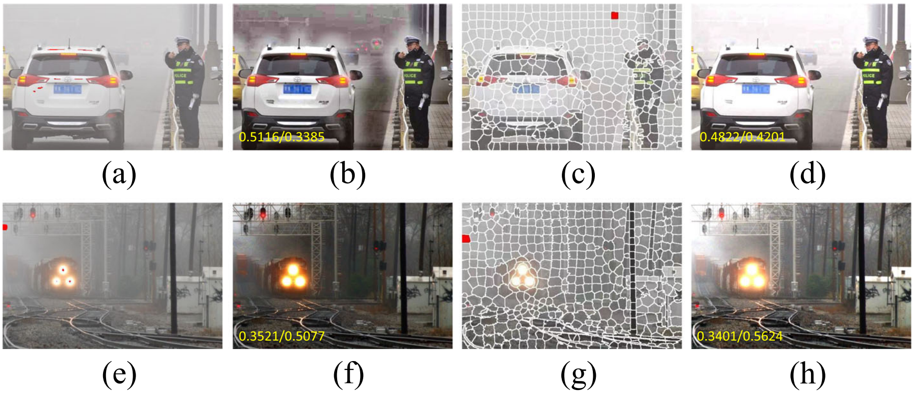

Fig. 4 Comparison of atmospheric light estimation methods. The atmospheric light is estimated to be obtained from the position shown in red. The DCP method is biased towards regions with bright objects (e.g. car, light), while our method is able to avoid these regions. (a) (e) The DCP method. (c) (g) The proposed method. (b) (f) The recovery images using the DCP method. (d) (h) The recovery images using the proposed method

The scene albedo is obtained by substituting the atmospheric light A^c and transmission map $t(x)$ into (1). Since the atmospheric light A^c has already been solved, our goal is to estimate the transmission map. (2) reveals that estimating the transmission map solely based on scene depth $d(x)$ is inadequate, the scattering coefficient β also plays a crucial role. Therefore, in order to obtain the transmission map, we designs an optimization function with the scattering coefficient β as the independent variable.

To calculate the value of β, we design an optimization function with the image quality-aware indicators as the constraint. It is expressed as:

$$\beta = \arg\min_{\beta} \left\{ f\left(dehaze\left(I^c(x), A^c, d(x), \beta \right), I^c(x) \right) \right\} \tag{14}$$

where $f(\cdot)$ represents an image quality-aware indicator. The image quality-aware indicator is formulated to maximize the texture information, minimize shift in the chrominance and minimize haze density of the restored image. Finally, the image quality-aware indicator is defined as:

$$\begin{aligned} f\left(Ot^c(y), \mathrm{In}^c(y)\right) = \xi \left(\sum_y \left(\sum_{c\in\{R,G,B\}} \nabla \left(Ot^c(y) \right) \right) \right) - \mathrm{var}\left(Ot_{MSCN}(y) \right) \\ + \lambda \left(\sum_y \left| \theta\left(\mathrm{In}^c(y) \right) - \theta\left(Ot^c(y) \right) \right| \right) \end{aligned} \tag{15}$$

where

$$\theta(y) = \cos^{-1} \left\{ \frac{\left[(y_R - y_G) + (y_R - y_B) \right]/2}{\sqrt{(y_R - y_G)^2 + (y_R - y_B)(y_G - y_B)}} \right\} \tag{16}$$

$$Ot_{MSCN}(y) = \frac{Ot_{gray}(y) - \mu(y)}{\sigma(y) + 1} \tag{17}$$

where

$$\mu(y) = \sum_{k=-K}^{K} \sum_{l=-L}^{L} \omega_{k,l} Ot_{gray}(y) \tag{18}$$

$$\sigma\left(y\right)=\sqrt{\sum_{k=-K}^{K}\sum_{l=-L}^{L}\omega_{k,l}\left[Ot_{gray}\left(y\right)-\mu\left(y\right)\right]^{2}} \tag{19}$$

where In is the hazy image, Ot is the recovery image, y denotes the coordinate of any pixel in In and Ot. $\nabla(\cdot)$ denotes the gradient operator, $\mathrm{var}(\cdot)$ the operator that calculates the variance. y_R denotes the red component of y, y_G is the green component of y, y_B is the blue component of y. $\omega=\left\{\omega_{k,l}\mid k=-K,\ldots,K,l=-L,\ldots,L\right\}$ is a 2D circularly symmetric gaussian weighting function sampled out to 3 standard deviations ($K=L=3$) and rescaled to unit volume, Ot_{gray} refers to the gray version of the image Ot.

The first term in (15) preserves the texture information in the recovered image Ot. For natural foggy images, the variance of the mean subtracted contrast normalized (MSCN) coefficients decreases with increasing haze density [10]. The second term in the equation utilizes the variance of the MSCN coefficients to estimate the haze density in the recovered image, while the third term computes the chrominance shift between the restored image Ot and the hazy input image In. To balance the contributions of these terms, the regulation parameters are designed as $\xi=-0.0001$ and $\lambda=0.01$. The scattering coefficient can be obtained when the image quality-aware indicator achieves the smallest value.

Note, the (15) is a $1D$ optimization function. Since the Fibonacci algorithm involves only addition and subtraction operations, it effectively reduces the complexity of the operation. In this work, the Fibonacci algorithm is adopted to solve (15). To solve the transmission map $t(x)$, scene depth $d(x)$ is required. In general, scene depth increases with haze density. Therefore, we assume that there is a positive correlation between haze density and scene depth. Inspired by Peng et al. [35], we use the minimum and maximum channels to calculate the scene depth, which is expressed as:

$$d(x)=I_{\min}(x)\left(1-\frac{I_{\max}(x)-I_{\min}(x)}{I_{\max}(x)}\right) \tag{20}$$

where $I_{\min}(x)=\underset{c\in\{R,G,B\}}{min}I^{c}\left(x\right)$ and $I_{\max}(x)=\underset{c\in\{R,G,B\}}{max}I^{c}\left(x\right)$.

Now, the atmospheric light A^c and transmission map $t(x)$ are determined, substituting (2) and (3) into (1), the scene albedo (i.e., haze-free image) ρ^c can be derived as:

$$\rho^{c}(x)=\frac{I^{c}(x)-A^{c}}{A^{c}\cdot e^{-\beta\cdot d(x)}}+1 \tag{21}$$

Since $\rho^c\in[0,1]$, the scene albedo is finally expressed as:

$$\begin{aligned}\rho^{c}(x)&=dehaze\left(I^{c}(x),A^{c},d(x),\beta\right)\\&=\min\left(\max\left(\left(\frac{I^{c}(x)-A^{c}}{A^{c}\cdot e^{-\beta\cdot d(x)}}\right)+1,0\right),1\right)\end{aligned} \tag{22}$$

where $dehaze(\cdot)$ represents the albedo restoring function, which takes four parameters, I^c is the input image, atmospheric light A^c can be calculated according to (13), $d(x)$ is the scene depth and β is the scattering coefficient. For clearer presentation, the entire process of scene albedo estimation for the proposed method is summarized in Algorithm 2.

Figure 5 shows dehazed results on example images with different β values. It can be observed that the dehazing and dim effect become stronger as β increase. When $\beta=1$, Fig. 5(d) has a good dehazing effect, but images become dim and lose details. When the β value is set to 1.5, Fig. 5(e) become dark in some areas and the objects in the areas are inconspicuous. We use adaptive atmospheric scattering coefficients in Fig. 5(f). It can be seen that the restoration results remove haze well and have no oversaturation.

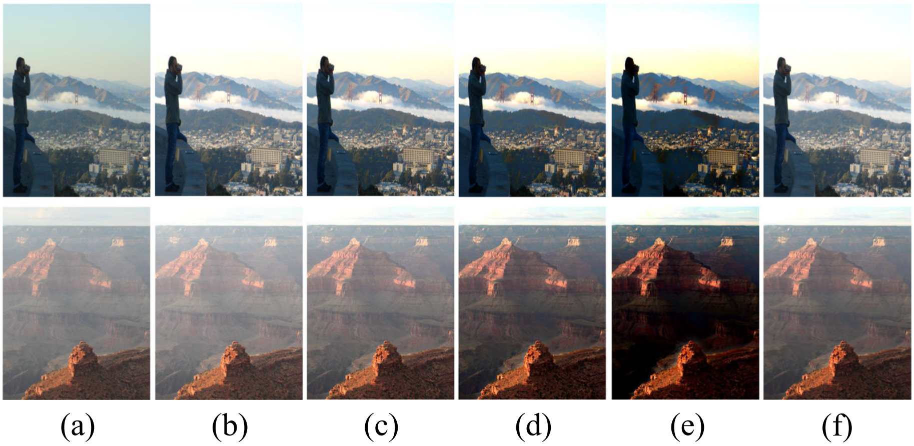

Fig. 5 Influence of different β values on the dehazing of the two images. (a) Original image. (b)-(e) The β values of 0.1, 0.5, 1, 1.5. (f) The β values are 0.3 and 0.6 respectively obtained using the proposed method

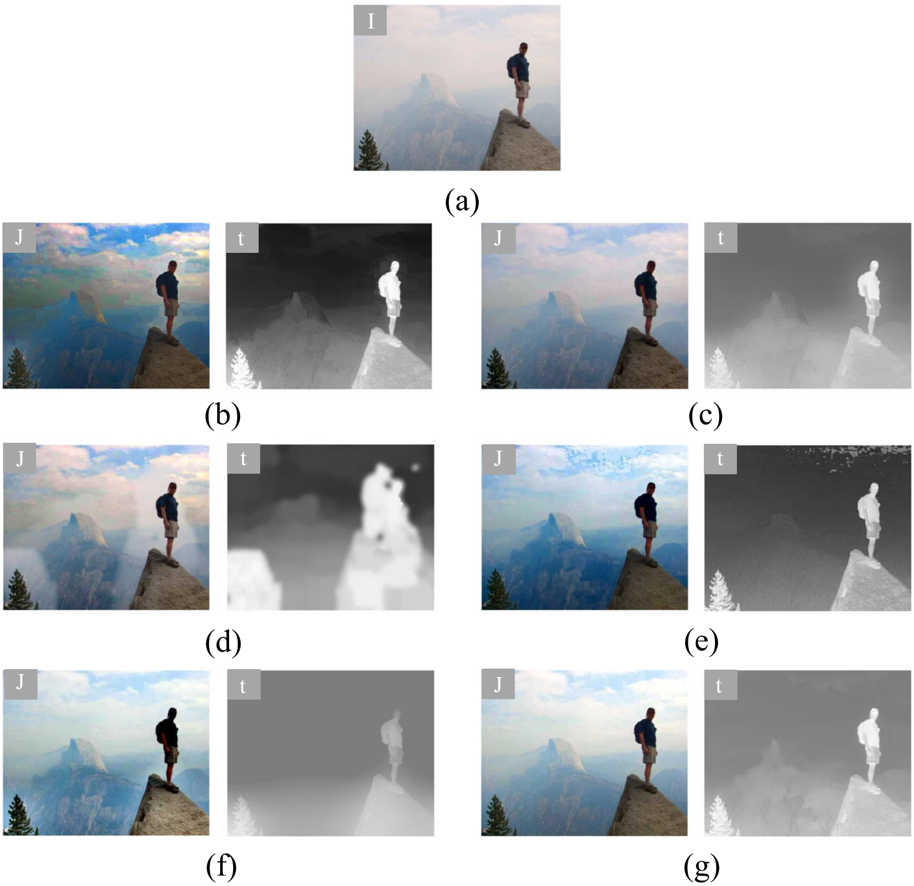

Fig. 6 Example of dehazed images and corresponding transmission maps using all comparisons methods. (a) Hazy image. (b) DCP. (c) DehazeNet. (d) PMS-Net. (e) IDGCP. (f) IDE. (g) Ours

Algorithm 2 Scene albedo ρ^c recovery algorithm

Input: original image I, atmospheric light value A^c, fibonacci algorithm search interval $[a, b] = [0, 5]$
Do the job:
1. Compute the scene depth $d(x)$ using (16)
2. Initialize the Fibonacci sequence Fib such that $\mathrm{Fib}[n] > b - a$
3. Define the interior points x_1 and x_2, set their initial values:
   $x_1 = a + \frac{Fib[n-2]}{Fib[n]} \times (b - a)$
   $x_2 = a + \frac{Fib[n-1]}{Fib[n]} \times (b - a)$
   do
   Compute O_{t1} and O_{t2} using (15):
   $O_{t1} = dehaze(I, A, d, x_1)$
   $O_{t2} = dehaze(I, A, d, x_2)$
   Calculate quality perception indicators f_1 and f_2 using (18)
   if $f_1 < f_2$ then
   Update the right endpoint of the interval to x_2, set x_2 to x_1, and compute x_1 and x_2 again.
   end
   if $f_1 \geq f_2$ then
   Update the left endpoint of the interval to x_1, set x_1 to x_2, and compute x_1 and x_2 again.
   end
   while $(x_2 - x_1) < eps$
   Set the midpoint of the interval as β, i.e. $\beta = \frac{x_1 + x_2}{2}$
4. Compute the Scene albedo ρ^c using (15)

Output: Scene albedo ρ^c

For model-based dehazing methods, the accuracy of transmission is critical to the dehazing effect. Figure 6 shows dehazed results and transmissions of the proposed method and the state-of-the-art methods, including DCP [18], DehazeNet [6], PMS-Net [41], IDGCP [22], and IDE [23].

As observed in Fig. 6, DCP refines the initial transmission with soft matting to mitigate the blocking artifacts. However, the transmission of the sky area is not improved, the edges and textures of people and trees are enhanced. The recovered image produced serious blocking artifacts in the sky and obvious artifacts around people and trees. The transmissions calculated by PMS-Net and IDGCP have darker sky regions. These methods produce artifacts and oversaturation in sky regions, as shown in Figs. 6(d) and 6(e). The transmissions obtained by Dehaze-Net and IDE are brighter, and the recovered result does not produce halo artifacts or color casts in the sky area. However, the pixel values of the human area in the dehazed image are significantly lower than those of the original image in the same area. In contrast, the transmission obtained by our method does not show blocking artifacts. In addition, the transmission not only reflects the depth variation between different objects but also improves the transmission values of the sky region.

4 Experimental results

In this section, we conduct experiments on the RTTS and FADE [10] datasets in this section. The RTTS dataset is from a large-scale synthetic haze dataset called RESIDE [31], which includes 4233 real-world haze images collected from the internet. FADE consists of 500 natural haze images and 100 testing images. We compare it with state-of-the-art image dehazing methods, including DCP [18], DehazeNet [6], PMS-Net [41], IDGCP [22], IDE [23], and AirNet [25]. Among these comparison methods, DCP, IDE, and IDGCP are prior-based dehazing methods, and the other three methods are based on deep neural networks.

Table 1 Comparison of dehazing results under different image patches N

image patches N	FADE	SSEQ	UCIQE	HRAM	Runtime
$N = 200$	1.0632	39.4105	0.54792	0.6330	3.672
$N = 300$	1.0874	39.4969	0.54351	0.6802	3.715
$N = 400$	1.0539	39.6048	0.54711	0.6806	3.774
$N = 500$	1.0291	39.7874	0.54882	0.6979	3.828
$N = 600$	1.0354	39.3664	0.54865	0.6837	3.874
$N = 700$	1.0132	39.8096	0.5473	0.6959	3.933
$N = 800$	0.9826	39.6496	0.5460	0.6945	3.995

These comparison methods are sourced from the open source code provided in the respective papers. All experiments were implemented on a PC with Intel(R) Core(Tm) i7–4790 CPU @ 3.60 GHz 16.0 GB RAM.

In addition, the quality of the restored images was evaluated using four metrics: spatial-spectral entropy-based quality(SSEQ) [33], underwater color image quality evaluation (UCIQE) [40], fog aware density evaluator (FADE) [10], and hue restoration ability metric (HRAM) [30]. The values of SSEQ range from 0 (worst) to 100 (best). UCIQE is a composite score that combines three metrics of underwater image quality: chroma, saturation, and contrast. Its range is from 0 to 1, and higher scores indicate better image quality. FADE quantifies the haze density of the dehazed image. A low FADE represents that the dehazed image has less mist residue. HRAM uses histogram similarity to measure the degree of hue shift in dehazed images. HRAM is similar to UCIQE, the higher, the better.

4.1 Parameter analysis

In this section, we analyze how to choose the image patches N. We select several hundred images from the dataset and process them using varying numbers of image patches ($N = 200, 300, \ldots, 800$) to obtain the dehazed images. Table1 displays the objective metrics and runtime comparison of different segmentation numbers on the dehazing results. Based on the overall dehazing performance and computational efficiency, we selected the number of image patches $N = 500$.

4.2 Qualitative assessment

To verify the effectiveness of the proposed method, a comparative analysis is conducted on four major scenarios: heavy haze, light haze, scenes with large sky areas, and complex background multi-object images. Figure 7 shows the dehazing results of different methods in heavy haze scenes. From Fig. 7(a), it can be seen that the haze distribution almost occupies the entire image. Observing the dehazing results of the DCP method, it can be found that the images overall suffer from darkening. The DehazeNet method leaves behind haze residue and exhibits poor generalization ability. The PMS-Net method exhibits halo artifacts at the edges. Most of the dehazed images produced by IDGCP are blurry, especially in the distant area. IDE method increases the overall brightness of each restored image but unfortunately introduces over-saturation. The restored images by AirNet suffer from significant brightness reduction

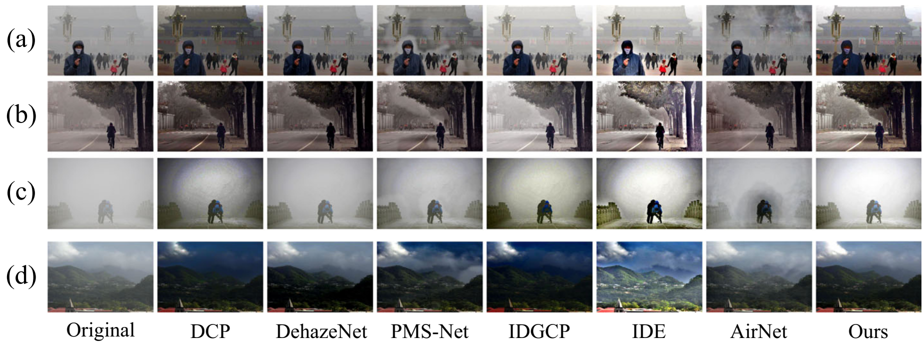

Fig. 7 Image dehazing results of seven different methods in heavy haze scenes

and loss of detail information. In contrast, the proposed method successfully removes heavy haze without introducing significant halo artifacts.

Figure 8 shows the dehazing results of different methods in light haze scenes. It can be seen from Fig. 8 that the visibility of the light haze images is higher compared to the heavy haze images and most of the detail information is retained. Taking the second row of images as an example, the dehazing results of the DCP method exhibit significant halo artifacts in the sky area. The DehazeNet method can effectively removes the haze but the dehazed images appear dim. The PMS-Net method produces obvious halo artifacts at the edges. Both the IDGCP and IDE methods result in overall brightness that is too high in the dehazed images. When dealing with colorful light haze images, slight oversaturation is present in the dehazing results of DCP and DehazeNet methods. The color restoration of PMS-Net is not as vivid as the proposed method. The dehazed results of IDGCP method have good color fidelity, but its overall dehazing ability is insufficient. The IDE method introduces color distortion. The dehazing results of AirNet method are close to that of the proposed method.

The results of different methods in scenes containing a large sky area are shown in Fig. 9. It can be seen from the Fig. 9 that the sky area in the dehazed results of DCP, IDGCP, and IDE methods exhibits noticeable color distortion. The dehazed results of DehazeNet method lose some detail information and color saturation. In the dehazed results of PMS-Net method, the edge structures appear blurry and produce artifacts. The results of AirNet method and the

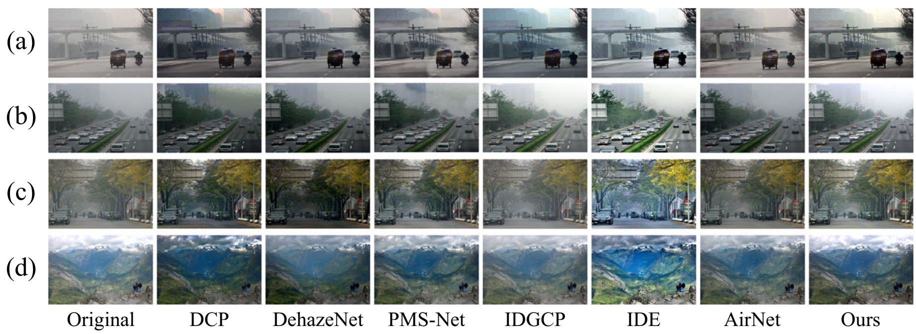

Fig. 8 Image dehazing results of seven different methods in light haze scenes

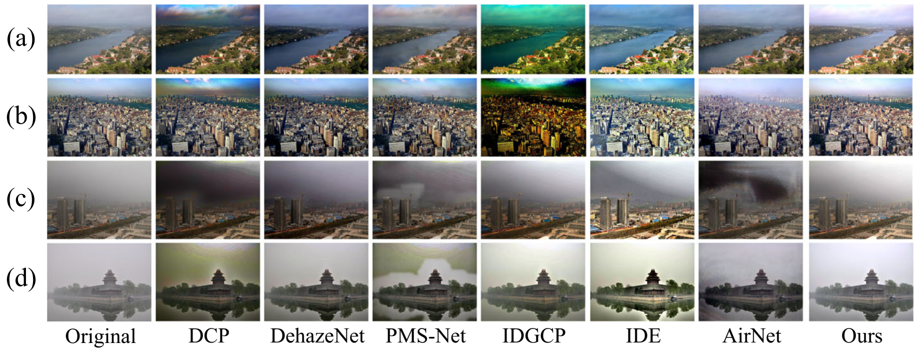

Fig. 9 The image dehazing results of seven different methods in scenes containing a large sky

proposed method in this paper do not exhibit color distortion in the sky area, but the detail restoration effect of AirNet method is slightly worse.

Figure 10 shows the dehazing results of different methods in scenes with cluttered background objects. It can be seen that DCP has a certain degree of dehazing effect, but the images appear overall dark due to inaccurate estimation of atmospheric light. DehazNet has a certain degree of haze residue. The results of PMS-Net method have good dehazing effect, but the edge parts are relatively blurry. The IDGCP method exhibits poor dehazing performance in real scenes, with generally dark images and slight color deviation. In order to enhance visibility, the IDE method applies exposure processing to the hazy image. However, this approach may lead to color distortion issues and unnatural visual perception due to overexposure. Nevertheless, due to its significantly improved overall visibility, it holds certain application value in specific scenes. The AirNet method exhibits limited generalization ability and shows some residual haze in the dehazing results. In comparison, the method proposed effectively removes haze while preserving natural color and detail information, aligning more closely with human visual perception.

Tables 2, 3, 4, and 5 display the quantitative evaluation results of seven methods for sixteen real-world hazy images selected in Sect. 4.2, with the last row in each table providing the average values. Table 6 presents the average objective metrics over the entire RTTS dataset. By analyzing the data in the tables, it can be observed that the IDGCP, IDE, and DCP methods perform well in terms of the FADE metric. However, the dehazed results of

Fig. 10 Image dehazing results of seven different methods in scenes with cluttered background objects

Table 2 Image dehazing evaluation based on SSEQ ↑

Image	DCP	DehazeNet	PMS-Net	IDGCP	IDE	AirNet	Ours
Heavy haze scenes							
Fig. 7(a)	49.9573	40.8801	41.3651	40.3759	43.0485	32.1746	50.8931
Fig. 7(b)	29.2256	27.0652	10.2658	29.2951	33.1394	5.2683	30.1251
Fig. 7(c)	0.2382	24.6104	2.1738	0.4290	1.0373	19.3250	8.9176
Fig. 7(d)	40.5216	37.9219	26.2209	38.8429	41.7463	34.6554	38.2254
Average	29.9856	32.6194	20.0064	27.2357	29.7428	22.8558	32.0403
Light haze scenes							
Fig. 8(a)	13.9581	32.8621	40.8205	35.6712	36.8166	25.7169	39.2068
Fig. 8(b)	6.0610	10.7674	13.9684	11.0063	10.3285	15.6820	10.6819
Fig. 8(c)	27.3203	24.0412	6.7692	22.4815	29.5805	13.0716	25.1938
Fig. 8(d)	25.6147	27.7592	8.7594	26.9286	16.9357	18.3829	18.7163
Average	18.2385	23.8574	17.5793	24.0219	23.4153	18.2133	23.4497
Scenes containing a large sky							
Fig. 9(a)	28.3854	32.6391	29.7982	33.8147	28.7923	13.7852	29.9842
Fig. 9(b)	19.5962	16.5475	13.7071	19.6502	19.2658	21.0187	19.2486
Fig. 9(c)	37.6675	37.0831	27.5296	33.9877	38.7471	35.6198	36.2797
Fig. 9(d)	40.0315	43.1305	34.0497	42.1373	41.2501	33.6283	35.7025
Average	31.4201	32.3525	26.2711	32.4424	32.0138	26.0130	30.3037
Scenes with cluttered background objects							
Fig. 10(a)	11.5862	17.8629	36.6291	6.8756	12.6548	10.8198	11.5474
Fig. 10(b)	32.1097	35.4802	24.5216	35.6405	45.2453	33.2189	46.9615
Fig. 10(c)	40.0315	43.1305	34.0467	42.3117	41.2501	33.6283	35.7029
Fig. 10(d)	41.8741	40.1774	18.5554	43.8821	44.5532	13.1728	42.2689
Average	31.4003	34.1627	28.4382	32.1774	35.9258	22.7099	34.1201

Table 3 Image dehazing evaluation based on UCIQE ↑

Image	DCP	DehazeNet	PMS-Net	IDGCP	IDE	AirNet	Ours
Heavy haze scene							
Fig. 7(a)	0.5586	0.5036	0.5015	0.5434	0.5715	0.4661	0.6063
Fig. 7(b)	0.5294	0.4815	0.5024	0.5182	0.5708	0.4823	0.6059
Fig. 7(c)	0.4635	0.3581	0.4316	0.4924	0.5603	0.4382	0.4295
Fig. 7(d)	0.5644	0.5716	0.6052	0.5937	0.6106	0.5473	0.6149
Average	0.5289	0.4787	0.5101	0.5369	0.5783	0.4834	0.5641
Light haze scenes							
Fig. 8(a)	0.5302	0.5006	0.5038	0.5609	0.5494	0.4772	0.5724
Fig. 8(b)	0.5061	0.4925	0.4706	0.5004	0.5506	0.4653	0.5168
Fig. 8(c)	0.6280	0.5730	0.5847	0.4962	0.6338	0.5619	0.5304
Fig. 8(d)	0.6149	0.5564	0.5661	0.5271	0.6351	0.5403	0.5437
Average	0.5698	0.5306	0.5313	0.5211	0.5922	0.5111	0.5408
Scenes containing a large sky							
Fig. 9(a)	0.6317	0.6228	0.5825	0.6705	0.6603	0.5827	0.6028
Fig. 9(b)	0.6528	0.6427	0.6234	0.7049	0.6685	0.6259	0.6259
Fig. 9(c)	0.5372	0.5290	0.5274	0.5731	0.6053	0.5104	0.5327
Fig. 9(d)	0.5040	0.4531	0.4825	0.4760	0.2584	0.4615	0.4594
Average	0.5814	0.5619	0.5539	0.6061	0.5481	0.5451	0.5552
Scenes with cluttered background objects							
Fig. 10(a)	0.6048	0.6034	0.6080	0.6112	0.6062	0.5666	0.5948
Fig. 10(b)	0.6313	0.5824	0.5733	0.5761	0.6574	0.5509	0.6217
Fig. 10(c)	0.5612	0.5530	0.5704	0.5641	0.5653	0.5234	0.5509
Fig. 10(d)	0.5162	0.4902	0.5091	0.5516	0.6234	0.4706	0.5706
Average	0.5783	0.5572	0.5652	0.5757	0.6130	0.5278	0.5845

Table 4 Image dehazing evaluation based on FADE ↓

Image	DCP	DehazeNet	PMS-Net	IDGCP	IDE	AirNet	Ours
Heavy haze scenes							
Fig. 7(a)	1.5318	1.7306	1.3509	2.3917	1.4530	1.0926	1.2761
Fig. 7(b)	0.4638	0.3564	0.4403	0.9254	0.4940	0.4037	0.4637
Fig. 7(c)	1.4430	3.3453	1.1591	1.3406	0.9735	3.7013	2.2538
Fig. 7(d)	0.8644	0.6130	0.5077	0.8444	0.6413	1.4308	1.1257
Average	1.0757	1.5113	0.8645	1.3755	0.8904	1.6571	1.2798
Light haze scenes							
Fig. 8(a)	0.8606	0.9901	1.1164	0.9530	0.8007	0.9513	0.7109
Fig. 8(b)	0.4452	0.4619	0.4973	0.7416	0.4532	0.5801	0.5237
Fig. 8(c)	0.2876	0.2189	0.3084	0.6429	0.2408	0.3219	0.2394
Fig. 8(d)	0.3609	0.4875	0.4736	0.7216	0.2640	0.4603	0.3208
Average	0.4885	0.5396	0.5989	0.7647	0.4396	0.5784	0.4487
Scenes containing a large sky							
Fig. 9(a)	0.4734	0.6149	0.7931	0.3641	0.4953	0.7361	0.5643
Fig. 9(b)	0.1894	0.1793	0.1954	0.1126	0.2179	0.2394	0.1860
Fig. 9(c)	1.1793	1.1052	0.8096	1.4607	0.9461	0.8572	1.6883
Fig. 9(d)	1.7741	3.3170	1.4749	3.8060	2.0276	1.7868	3.4146
Average	0.9040	1.3041	0.8182	1.4358	0.9217	0.9048	1.4633
Scenes with cluttered background objects							
Fig. 10(a)	0.2172	0.2031	0.2989	0.2522	0.2188	0.2779	0.2658
Fig. 10(b)	0.6003	0.7074	0.6713	1.0268	0.6333	0.6951	0.8535
Fig. 10(c)	1.7741	3.3170	0.5094	3.8060	2.0276	1.7868	1.1064
Fig. 10(d)	0.9062	1.5325	1.0247	1.4209	0.8614	0.7560	0.9816
Average	0.8779	1.4400	0.6260	1.6264	0.9352	0.8789	0.8018

Table 5 Image dehazing evaluation based on HRAM ↑

Image	DCP	DehazeNet	PMS-Net	IDGCP	IDE	AirNet	Ours
Heavy haze scenes							
Fig. 7(a)	0.4206	0.5091	0.4401	0.3917	0.4962	0.4408	0.5314
Fig. 7(b)	0.6531	0.3641	0.6671	0.5506	0.7038	0.6423	0.6704
Fig. 7(c)	0.3764	0.5816	0.4380	0.4973	0.3715	0.5936	0.4791
Fig. 7(d)	0.5613	0.3690	0.6144	0.5221	0.5186	0.5846	0.5975
Average	0.5028	0.4559	0.5399	0.4904	0.5225	0.5653	0.5696
Light haze scenes							
Fig. 8(a)	0.6794	0.7469	0.7516	0.7023	0.7649	0.8004	0.7932
Fig. 8(b)	0.7193	0.7461	0.7931	0.8192	0.7204	0.8361	0.8801
Fig. 8(c)	0.6413	0.6034	0.8425	0.9903	0.8713	0.8802	0.8433
Fig. 8(d)	0.4628	0.6915	0.7139	0.9907	0.8806	0.8134	0.8209
Average	0.6257	0.6969	0.7752	0.8756	0.8093	0.8325	0.8343
Scenes containing a large sky							
Fig. 9(a)	0.6314	0.8371	0.7830	0.7264	0.6314	0.8627	0.7706
Fig. 9(b)	0.8827	0.8290	0.8912	0.5374	0.6307	0.9403	0.8137
Fig. 9(c)	0.6092	0.7547	0.7062	0.7199	0.6752	0.6420	0.8153
Fig. 9(d)	0.5529	0.6980	0.5925	0.4861	0.4985	0.7049	0.4609
Average	0.6746	0.7797	0.7432	0.6424	0.6089	0.7874	0.7651
Scenes with cluttered background objects							
Fig. 10(a)	0.6467	0.6097	0.7046	0.6293	0.6307	0.7972	0.8092
Fig. 10(b)	0.4290	0.5222	0.7040	0.5799	0.3748	0.5854	0.5348
Fig. 10(c)	0.5529	0.6980	0.6225	0.4861	0.4985	0.6049	0.5409
Fig. 10(d)	0.3354	0.3561	0.4028	0.5604	0.6108	0.4253	0.5723
Average	0.4910	0.5456	0.6084	0.5639	0.5287	0.6032	0.6143

Table 6 Comparison of average objective evaluation results on the RTTS dataset

Image	DCP	DehazeNet	PMS-Net	IDGCP	IDE	AirNet	Ours
FADE	0.9988	1.1675	1.2858	0.6568	0.9217	1.2476	1.0835
UCIQE	0.5666	0.5253	0.5291	0.5476	0.5675	0.5182	0.5478
SSEQ	37.2711	38.0820	33.7757	39.0926	39.2115	30.6501	38.1008
HRAM	0.5859	0.5530	0.6200	0.5732	0.5905	0.5675	0.6515

the DCP method exhibit obvious color casts in the sky region, while both the IDGCP and IDE methods lead to over-brightened dehazed images, indicating their potential value in certain scenarios. The proposed method achieves top-three results in SSEQ, UCIQE, and HRAM metrics, demonstrating its competitiveness, especially its best performance in the HRAM metric, further proving its outstanding ability in restoring the color information of the dehazed images.

4.3 Subjective evaluation

To further illustrate the validity of the experimental results, ten observers without visual problems completed the subjective evaluation of this section. Note that none of the authors of this article participated in this subjective study. Firstly, the experimenters showed the observers ten real haze-free images to let the participants understand what the haze-free images look like. After that, the four scenarios from Sect. 4.2 were presented to the observers in the form of 15 images. They were asked to rate these images (0–5). Scoring was based on the presence of color casts and halo artifacts in the image, whether the image looked natural, and whether the image had good contrast.

Figure 11 shows the subjective evaluation results, indicating that the observers believe that the dehazed images obtained by our method are better. The proposed method achieved a high score due to its superior performance in enhancing the contrast and saturation of the dehazed images, making them more visually appealing and perceptually suitable for humans.

4.4 Influence of image size on calculation results

In this section, experiments are conducted to evaluate the runtime of the proposed method and other comparative methods at different image sizes. Specifically, DCP, DehazeNet, IDGCP, IDE, and the proposed method are tested in Matlab R2020a software on a computer with an Intel(R) Core(Tm) i7–4790 CPU @ 3.60 GHz. The AirNet method is implemented using PyTorch in publicly available source code, and therefore, its runtime is tested on an Nvidia GeForce RTX 3090 GPU. It should be noted that the input image size for PMS-Net in the publicly available model is limited to 640 $\times$ 480, and therefore, it is not included in these experiments.

Figure 12 shows the corresponding running times for the DCP, DehazeNet, IDGCP, IDE, AirNet and the proposed method. The AirNet method has significantly faster computation time than other algorithms, thanks to its use of GPU parallel computing. However, it is challenging to use on low computing power platforms. The proposed method has slightly higher computational complexity compared to IDGCP. IDGCP achieves high-quality restoration by maximizing the texture information of the evaluated image while minimizing information loss

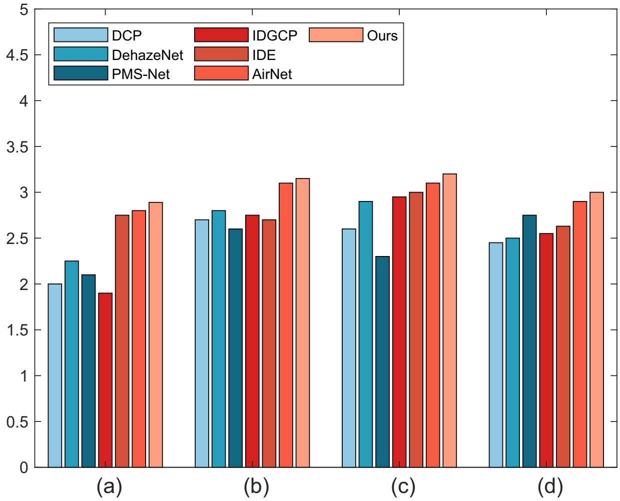

Fig. 11 Subjective evaluation on haze images of four different scenarios. (a) The average subjective evaluation results in the heavy haze scenes. (b) The average subjective evaluation results in the light haze scenes. (c) The average subjective evaluation results for scenes containing a large sky. (d) The average subjective evaluation results for scenes with cluttered background objects

through the design of a visual indicator based on single image priors. The proposed method designs an optimization function with image quality-aware indicators to achieve dehazing while preserving true colors. The image quality-aware indicator is formulated to maximize texture information, minimize chrominance shift, and minimize haze density in the restored image. Although both IDGCP and the proposed method are optimization approaches, the optimization function for the proposed method is relatively complex, resulting in longer runtime. DehazeNet, IDE, and the proposed method exhibit similar computational complexity and runtime.

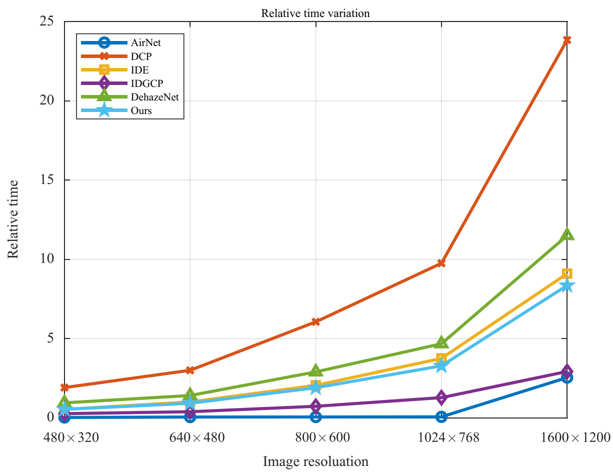

Fig. 12 Computation efficiency comparison

5 Conclusion

In this paper, we propose an image quality-aware approach with adaptive scattering coefficients for image dehazing. First, we find the most likely candidate for atmospheric light by the regional rank-based method. Then, we design an optimization function constrained by the image quality-aware indicators to compute the scattering coefficient or transmission. The optimization problem is solved using the Fibonacci algorithm. Finally, the dehazed image is calculated by putting the atmospheric light and transmission into an optical model. Benefitting from the image quality-aware indicators, our method can make the dehazed results have high contrast and rich texture details. However, the limitation of the proposed method is the relatively long computation time. In future research, we will focus on optimizing the algorithm to reduce the computation time, making it more practical.

Data Availability Statement The datasets generated and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflicts of interest The authors declare that they have no potential conflict of interest

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